Dataset and Baseline Methods for Structure Analysis of Variations on Chinese Folk Songs

Anonymous ICME submission

***Abstract*—Chinese folk song variation pieces represent a musical structural form that draws on Western variation theory while integrating distinctive Chinese ethnic culture and aesthetic principles. Despite the rapid advancement of music structure analysis driven by neural networks, research on the structural analysis of Chinese folk song variations remains limited, particularly due to the lack of systematic and large-scale annotated datasets. To address this gap, we construct a structural analysis dataset consisting of 553 Chinese folk song variation pieces (CFVSD) and propose the SA-CNFolk model based on this dataset. The proposed model achieves an HR3F score of up to 82.41% in the structural boundary prediction task, demonstrating its effectiveness in modeling the structural characteristics of Chinese folk song variations. Overall, this work represents the first systematic effort to introduce and characterize Chinese folk song variations within the framework of computational music structure analysis, laying a foundation for cross-cultural music structure modeling.**

***Index Terms*—music structure, Chinese folk song, CFVSD, SA-CNFolk, variation form**

I. Introduction

Variation is a common and widely used compositional technique. Both music and literary rhetoric employ variation as an expressive device. Fundamentally, variation is built upon repetition, which in both music and literature primarily serves to emphasize particular ideas or emotions. Broadly speaking, variation functions as a developmental technique formed through the addition of various forms of elaboration and ornamentation, encompassing nearly all compositional techniques in music. Repetition with variation represents one of the earliest and most universal methods of musical development worldwide and occupies a central position in compositional practice. In contexts such as improvised strophic songs or dance music, variation has also been widely applied [1].

Western classical variation forms gradually emerged from this principle [1]. Over the course of their historical development, they became increasingly detached from literary and rhetorical functions, evolving into formalized musical structures centered on a theme and its subsequent transformations.

Chinese folk song variation pieces adopt conceptual elements from Western classical variation forms. However, due to fundamental differences in musical culture, Chinese folk song variations exhibit distinctive characteristics that are closely related to linguistic features and intrinsic musical structures. Based on an examination of the conceptual differences between Chinese folk song variations and Western variation forms, this paper conducts a systematic investigation into the structural organization of Chinese folk song variations.

However, in the field of music structure analysis, the majority of existing studies primarily focus on popular music. Early research largely relied on repetition-based characteristics between structural segments [2], [3]. Over time, researchers have recognized the inherent limitations of approaches that depend solely on repetition information. With the advancement of artificial intelligence, recent structure analysis methods have begun to incorporate deep learning models such as convolutional neural networks (CNNs) and Transformers. Based on the Harmonix Set [4], methods such as All-In-One [5] and the CBM algorithm [6] have achieved significant performance improvements under commonly used tolerance-based evaluation metrics (e.g., time tolerances measured in seconds).

Nevertheless, the defining characteristics of Chinese folk song variations lie in their hierarchical structural organization and the similarity among different variation sections. In other words, each variation section typically contains an internal structure composed of shorter musical phrases, while different variations often retain a certain degree of similarity in terms of thematic material. In contrast, such characteristics are relatively simpler in popular music. Therefore, whether structure analysis methods originally designed for popular music are sufficiently robust when confronted with the more complex structures of folk song variations remains an open question and warrants further investigation.

To enable structural analysis of Chinese folk song variations, this paper constructs and releases a dataset comprising 553 works, referred to as CFVSD. The dataset provides detailed annotations of structural information as well as regional metadata. All annotations and verification procedures were carried out by a team of faculty members and students with professional training in musicology, ensuring the accuracy and reliability of the annotated results. As the first dataset to include structural annotations for Chinese folk song variation pieces, CFVSD is expected to offer valuable support for structural research and practical applications in related fields, and to further promote the development of analytical and computational studies on Chinese folk song variations.

Based on the constructed CFVSD dataset, we further propose the SA-CNFolk model for structural analysis of Chinese folk song variations. The model consists of two main components: a feature embedding module and a feature aggregation module, implemented using a convolutional neural network (CNN) and a long short-term memory (LSTM) network, respectively. The feature embedding module is designed to extract local time–frequency representations, while the feature aggregation module models contextual dependencies between

TABLE I
META-INFORMATION OF THE WORLD FOLK SONG DATABASES

Dataset	Content	# of pieces
Cantocore [7]	Analyze the stylistic characteristics of folk songs from different geographical regions based on rhythm, pitch, and timbre.	259
Traditional songs from Japan and East Asia [8]	The similarity of stylized songs in music is analogous to the similarity of DNA type distributions.	680
Billboars [9]	Beat alignment and chord transcription	1379
Million Song [10]	Collect songs, including audio features related to pitch, timbre, and loudness.	24941
European Folksong [11]	Divide European folk songs into geographical regions based on their musical features.	950
Dutch folk songs [12]	Analyze the melodic similarity of Dutch folk songs.	360

musical segments as well as global structural information.

At the output stage, we further introduce a post-processing strategy to smooth and filter the predicted results, enabling the extraction of the most salient structural boundaries for each piece as the final structural prediction. Experimental results demonstrate that SA-CNFolk achieves superior performance in the structural boundary prediction task, reaching an HR3F score of up to 84.50% under a $\pm$3-second tolerance setting. Comparative experiments further show that this result represents an improvement of approximately 50% over existing mainstream structure analysis methods. Moreover, even under a more stringent tolerance setting ($\pm$0.5 seconds), SA-CNFolk maintains its leading performance, achieving the best HR.5F score to date, which further highlights its effectiveness in analyzing the structural characteristics of Chinese folk song variations. Overall, the main contributions of this paper are as follows:

- Constructed and released CFVSD[1], the first dataset of Chinese folk song variations with detailed structural annotations.
- Proposed SA-CNFolk, a structural analysis method capable of effectively handling the complex structural characteristics of Chinese folk song variations.
- Through systematic comparative experiments, revealed the limitations of existing popular music structure analysis methods on Chinese folk song variations, providing empirical evidence for cross-cultural music structure modeling research.

The remainder of this paper is organized as follows. Section II reviews existing folk song datasets and methods in the field of music structure analysis (MSA). Section III introduces Chinese folk song variations and the CFVSD dataset, including the specific challenges encountered during dataset collection. Finally, our proposed SA-CNFolk framework and the corresponding experiments are presented in Section IV and Section V, respectively.

II. LITERATURE REVIEW

In this section, we review datasets related to folk songs and discuss existing methods in the field of music structure analysis (MSA).

A. Dataset

Since the beginning of the twentieth century, computers have assisted researchers in storing and processing musical information, which in turn enabled systematic comparisons among traditional music from various regions [13]. With the continued development of the computational study of music, a number of datasets related to folk music have been accumulated to date. For example, Cantocore [7], Traditional Songs from Japan and East Asia [8], Billboars [9], MSD (Million Song Dataset) [10], European Folksong [11] and Dutch Folk Songs [12], to name a few. The meta-information of these datasets is summarized in Table I. Although recent research has begun to focus on folk songs from various regions around the world, there still exists no dataset specifically addressing musical structure, nor any dataset dedicated to Chinese folk song variations. Therefore, the present dataset focuses on Chinese folk song variations and is designed to support the modeling and analysis of their structural characteristics.

B. Method

With the continuous development of the music information retrieval (MIR) field, research on Chinese folk songs has gradually attracted attention. For instance, some studies have constructed datasets for emotion analysis of Chinese folk songs [14] to facilitate research in music emotion recognition. In addition, other researchers have explored the similarities and differences among Chinese folk songs from different regions [15]. However, existing datasets remain insufficient for structural analysis of Chinese folk songs. Therefore, the present study can be regarded as a significant step toward exploring the structure of Chinese folk song variations. Given that the structural analysis of popular music has become a widely studied area, this section first reviews and discusses the methods commonly used in popular music structure analysis.

Early methods for popular music structure analysis primarily relied on the computation of similarities between repeated segments. For example, Paulus and Klapuri [2] used a greedy search strategy to match repeated segments, while Shiu et al. [3] constructed similarity matrices to detect structural boundaries. However, repetitions in music often involve some

[1]The dataset link will be made publicly available upon acceptance of this paper.

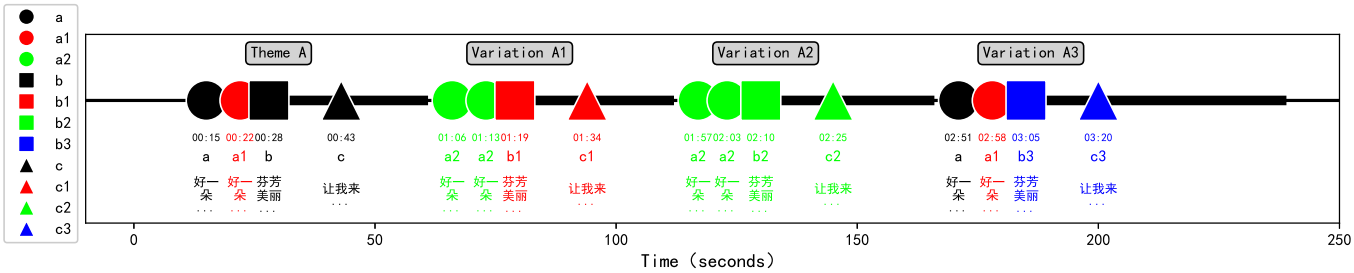

Fig. 1. Structure Diagram of Chinese Folk Song Variations

degree of variation, and reliance solely on similarity measures is limited in identifying complex structures. To address this limitation, researchers have increasingly employed artificial intelligence methods to automatically learn higher-level musical features from data, thereby improving the accuracy and robustness of structural analysis. For instance, Ullrich et al. [16] proposed a convolutional neural network (CNN) based model that incorporates novelty features representing significant transitional changes between musical segments to locate structural boundaries. Wang et al. [17] further combined CNNs and Transformers to predict the probability of each time point being a structural boundary and applied a local averaging post-processing strategy to extract probability peaks as final boundaries.

More recently, the All-In-One model proposed by Kim and Nam [5] achieved significant progress on the Harmonix Set dataset [4]. This method separates different instrument tracks and introduces a dilated attention mechanism, showing superior performance in boundary detection tasks. Buisson et al. [18] constructed pairwise links between musical segments and employed a graph attention mechanism to determine whether two segments belong to the same structural unit. Although this approach improved performance in HR.5F, which is the F1 score for structural boundary detection within a ± 0.5 second tolerance, its performance in HR3F, the F1 score within a ± 3 second tolerance, did not surpass the dynamic programming and self-similarity matrix based method proposed by Marmoret et al. [6].

In summary, although there is currently no dataset specifically designed for structural analysis of Chinese folk song variations, to comprehensively evaluate the effectiveness of our proposed method, we compare it with state-of-the-art approaches that have demonstrated strong performance in popular music structure analysis.

III. CFVSD Dataset

The structure of traditional Chinese music often integrates music with text, and the musical structure of folk songs is influenced by lyrics, resulting in variations that combine melody and semantic content, commonly referred to as *Ostinato Variation*. The example in the figure uses the Chinese folk song variation "Jasmine Flower" to illustrate the discussion. Through this example, it can be observed that Chinese folk song variations integrate both musical and literary rhetorical development techniques, exhibiting certain differences compared to Western classical variation forms. "Jasmine Flower" is a widely circulated Chinese folk song, with its origins traceable to the Qing Dynasty melody "Xianhua Diao". Various regions in China have adapted the song to reflect local musical characteristics; in this study, we adopt the Jiangsu version of "Jasmine Flower".

The variation structure of "Jasmine Flower" is characterized by thematic development, consisting of a theme section A and three variation sections, A1, A2, and A3. Each section contains four musical phrases, labeled a, a1, b, and c. In this piece, the lyrics remain unchanged, and the variations are fully expressed through melody, pitch, and rhythm. In theme A, the first and second phrases (a and a1) undergo rhythmic changes in sections A1 and A2, and return to the rhythm of A1 in section A3. The third and fourth phrases of theme A exhibit changes in both rhythm and pitch in each variation section, without any repetition. In general, the variation techniques employed in Chinese folk songs are diverse, and there is no fixed pattern governing their application.

During dataset curation, the preliminary identification was primarily based on melody and lyrics. Some folk songs feature repeated lines of lyrics within a segment, while others exhibit minimal melodic variation, corresponding to constant-melody or cantus firmus variations, where the basic melody or principal tune remains largely unchanged. For these cases, the segmentation of variation sections relies heavily on the lyrical content. During the data collection process, we also encountered many songs with identical melodies but completely different titles and lyrics. Therefore, our main criterion for identifying variations is melodic analysis, with lyrics serving as a selective reference.

In summary, the CFVSD dataset will be publicly released[1]. It contains 553 Chinese folk song variations, with audio sourced from publicly available online resources. The annotators and verifiers are a team of students and faculty with professional music backgrounds. To ensure consistency in annotation, all participants underwent standardized training on the labeling process. The procedure is as follows: first, folk songs with variation structures are selected, and the corresponding audio files are downloaded from online platforms. Next, the annotators manually label metadata, including song

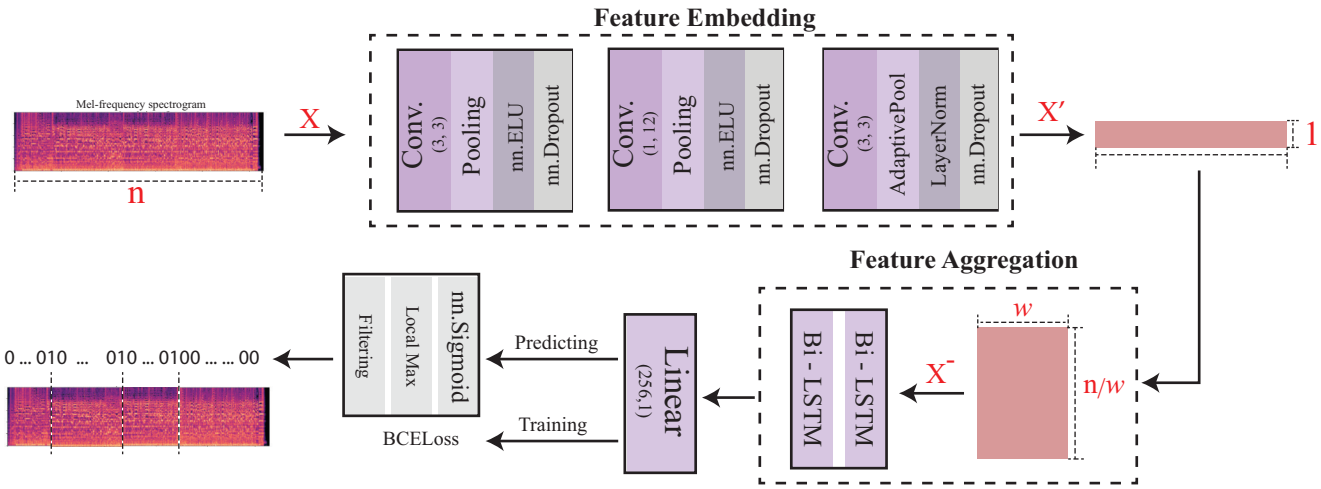

Fig. 2. Visualization of the SA-CNFolk framework.

title, performer, region, and album information. Meanwhile, the annotators will label the structure boundary. Simply put, the theme section, A, is first identified by annotators, and subsequent variations (A1, A2, A3, etc.) are labeled along with their respective starting times.

IV. SA-CNFolk Framework

Based on our curated CFVSD dataset, we propose a model named SA-CNFolk (Structure Analysis for Chinese Folk Music). The model is primarily composed of a Convolutional Neural Network (CNN) and a Long Short-Term Memory network (LSTM). Its overall architecture is illustrated in Figure 2.

Specifically, we first convert the input audio signal into a Mel-spectrogram, denoted as $X \in \mathbb{R}^{n \times f}$, where n represents the number of time frames and f is the number of Mel frequency bins. The Mel-spectrogram is a commonly used feature for music structure analysis tasks [15]. Next, we perform feature embedding on the Mel-spectrograms. High-level abstract features of the audio are extracted through a multi-layer convolutional neural network (CNN). In this process, the original single-channel input is mapped to a multi-channel feature representation, where the number of channels is controlled by a hyperparameter σ. Additionally, an adaptive pooling mechanism is introduced at the end of the last CNN layer to compress the feature dimension to 1. After these operations, the feature of each frame is represented as a σ-dimensional vector (i.e., with C channels), denoted as $X' \in \mathbb{R}^{n \times \sigma}$.

The subsequent feature aggregation is another key component of the model. In this stage, we first perform a reshape operation on the features. Specifically, given a window size w, the feature sequence X' output from the previous stage is segmented into non-overlapping windows of size w. Within each window, the features are flattened. The resulting feature matrix after segmentation and reorganization is denoted as $\hat{X} \in \mathbb{R}^{(w \times \sigma) \times \frac{n}{w}}$. This feature reorganization reduces the temporal resolution, encouraging the model to learn structural-level features rather than focusing solely on the details of individual frames. In other words, in our experiments, without feature reorganization, the number of frames per second (FPS) can be calculated as FPS $= \frac{\text{sampling rate}}{\text{hop length}} = \frac{44100}{512} \approx 86$ frames. Within a tolerance of ± 3 seconds, the SA-CNFolk model is required to ensure the classification results for the majority of $86 \times 6 = 516$ boundary frames, that is, most frames within this tolerance are predicted as class 1 (boundary frames). However, after feature aggregation, within a tolerance of ± 3 seconds, the model only needs to predict approximately $\frac{516}{w}$ frames as boundary frames for each boundary. Based on the reorganized feature matrix $\hat{X}$, we feed it into an LSTM layer for modeling. The LSTM captures temporal dependencies along the time dimension (i.e., $\frac{n}{w}$ dimension) and maps the $w \times \sigma$ features to a h-dimensional representation. It should be noted that we employ a bidirectional LSTM (Bi-LSTM) in this stage. Music, as a typical sequential data type, exhibits strong contextual dependencies in its structure. Therefore, the Bi-LSTM is able to effectively capture both forward and backward temporal information, thereby improving the accuracy of boundary detection.

Finally, after ℓ layers of Bi-LSTM for feature aggregation, the model predicts logits through a linear layer, followed by a Sigmoid activation for normalization. It should be noted that due to the temporal aggregation performed on the feature matrix, the time dimension is reduced from n to $\frac{n}{w}$, and thus the frame-level labels must be correspondingly adjusted. Specifically, for the original binary label sequence, where 0 represents non-boundary frames and 1 represents boundary frames, we denote it as $y = [y_1, y_2, \ldots, y_n]$. Due to our feature aggregation operation, we also apply a non-overlapping sliding window of size w to the label sequence y for aggregation. The aggregated label sequence is denoted as $\hat{y}$, where each aggregated label is computed as

$$\hat{y}_i = \frac{1}{w} \sum_{j=(i-1)w+1}^{iw} y_j, i = 1, 2, \ldots, \frac{n}{w}.$$

TABLE II
COMPARISON STUDY OF THE SA-CNFOLK FRAMEORK

Method	HR.5P (%)	HR.5R (%)	HR.5F (%)	HR3P (%)	HR3R (%)	HR3F (%)
All-In-One [5]	1.70	100.00	3.34	1.70	100.00	3.34
All-In-One [5] w/ Agg.	15.34	100.00	26.60	17.01	95.34	28.87
CBM [6]	15.69	47.92	23.64	23.33	61.94	33.89
SA-CNFolk	54.37	49.95	**52.07**	94.32	73.18	**82.41**

Here, the index j refers to the frame number in the original label sequence y. This operation converts the original binary 0-1 label sequence into a continuous probability sequence by computing the average within each window.

V. EXPERIMENT

A. Experiment Setting

Based on the CFVSD dataset contributed in this work, which contains structural annotations for Chinese folk song variations, we split the dataset into training, validation, and test sets with an 8:1:1 ratio. Since the dataset includes multiple renditions of the same folk song, we ensured that different versions of the same song were assigned to the same subset to prevent information leakage from affecting model evaluation. Moreover, the model that achieved the best performance on the validation set in terms of the HR3F metric was preserved for subsequent experiments and analysis. We also employed an early stopping strategy: training was terminated if the model's performance on the validation set did not improve for 5 consecutive epochs, in order to prevent overfitting.

B. Results and Analysis

To comprehensively evaluate the proposed SA-CNFolk model, we conducted hyperparameter tuning, ablation studies, and comparison experiments to verify the effectiveness of individual components as well as the overall performance. During hyperparameter tuning, we mainly focused on the following parameters: the feature aggregation window size w (candidate values: $0.25 \times FPS$, $0.5 \times FPS$,$1 \times FPS$), the number of channels in the feature embedding module σ (candidate values: 6, 12, 24), the hidden dimension of the LSTM layers (candidate values:64, 128, 256), and the number of LSTM layers ℓ (candidate values: 1, 2, 3).

The top five hyperparameter configurations achieving the best performance are shown in Table III. The evaluation metrics include Recall, Precision, and F1 score, all calculated under a $\pm$3-second tolerance and denoted as HR3R, HR3P, and HR3F, respectively. These metrics are widely used in music structure analysis tasks [14], [15]. The experimental results show that the best-performing hyperparameter combination is $w = 1.0, \sigma = 24, h = 128, \ell = 2$, achieving a highest HR3F score of 82.41%. Moreover, the HR3P reaches 94.32%, indicating that almost predicted boundary points are correct.

Based on the optimal hyperparameter configuration presented in Table III, we further conducted ablation studies to analyze the contributions of three key modules in the model, including the Feature Aggregation module, the LSTM layer within the aggregation module, and the Feature Embedding module. Specifically, when the Feature Aggregation module is entirely removed, the model degenerates to directly classifying high time-resolution frame-level features.

TABLE III
TOP 5 RESULTS FROM HYPER-PARAMETER TUNING OF THE SA-CNFOLK FRAMEWORK

(w, σ, h, ℓ)	HR3P (%)	HR3R (%)	HR3F (%)
$(1.0, 24, 128, 2)$	94.32	73.18	**82.41**
$(0.25, 24, 64, 2)$	94.17	73.02	82.26
$(1.0, 24, 128, 2)$	89.57	75.40	81.87
$(0.5, 12, 64, 3)$	94.93	71.91	81.83
$(1.0, 24, 256, 3)$	92.03	70.89	80.14

TABLE IV
ABLATION STUDY OF THE SA-CNFOLK FRAMEWORK

Method	HR3P (%)	HR3R (%)	HR3F (%)
w/o LSTM	14.03	100.00	24.61
w/o Agg.	3.17	100.00	6.14
w/o Emb.	100.00	40.62	57.77
SA-CNFolk	**94.32**	**73.18**	**82.41**

As shown in the results in Table IV, the Feature Aggregation module plays a crucial role in the model's performance. Without feature aggregation (i.e., the "- w/o Agg" row in Table IV), the high time-resolution predictions almost misclassify all frames as boundary frames. In fact, the determination of boundary frames usually relies on contextual information from several seconds before and after, meaning that short-term fluctuations at high time resolution are not essential. Nevertheless, even with feature aggregation, removing the LSTM layer still results in poor performance. This is mainly because, without the LSTM layer, the features are directly fed into a linear layer for classification, which lacks the capability to capture contextual dependencies.

Furthermore, in terms of evaluation metrics, we additionally introduced a stricter ±0.5-second tolerance, denoted as HR.5P, HR.5R, and HR.5F. The experimental results (see Table II) show that although SA-CNFolk achieves an excellent HR3F score of 82.40%, its performance drops to 30.34% under the more stringent HR.5F metric. This indicates that there is still room for improvement in the temporal precision of boundary prediction.

In addition, we compared our method with current mainstream approaches for pop music structure analysis, including AllInOne [5] and CBM [6]. As shown in Table II, although

these baseline models perform well on pop music structure analysis tasks, they exhibit significant limitations when applied to the structural segmentation of Chinese traditional variation songs. This is mainly because Chinese folk song variations maintain a certain degree of thematic consistency across different sections, and each variation segment often contains a hierarchical structure composed of shorter musical phrases.

Even when the baseline models are combined with the most critical Feature Aggregation module from SA-CNFolk (as in the "AllInOne [5] - w/ Agg." row in Table II), the resulting HR3F score reaches only about 28.87%. Similarly, the CBM algorithm [6] implements a comparable feature aggregation mechanism based on bar-level frames. However, as shown in Table [**?**], the performance of "AllInOne [5] - w/ Agg." and "CBM [6]" is similar, and under the relaxed tolerance setting (HR3F), their overall performance remains 48% lower than that of the SA-CNFolk method.

Overall, SA-CNFolk outperforms the existing baseline methods in terms of the HR3F metric, particularly under the 3-second tolerance setting, demonstrating its capability to accurately model boundary information. Meanwhile, the ablation studies reveal the critical contributions of both the Feature Aggregation module and the Bi-LSTM to the overall performance. Therefore, SA-CNFolk effectively captures the contextual dependencies of structural boundaries and serves as a robust method for music structure segmentation tasks in Chinese traditional variation songs.

VI. Conclusion

Given the importance of Chinese folk songs in traditional Chinese music, this paper introduces a variation-focused dataset centered on music structure research. The dataset comprises 553 folk songs with manually annotated variation structures, aiming to address the lack of music-structure-focused datasets in the folk music domain. Based on this dataset, we propose, for the first time, a model for analyzing the structural characteristics of Chinese folk song variations: the SA-CNFolk model (Structure Analysis of Chinese Folk Music). Under a second-level tolerance setting, the model achieves an HR3F score of 82.41%, representing a significant breakthrough in this field.

Even after retraining and hyperparameter tuning existing methods originally designed for pop music, SA-CNFolk demonstrates clear advantages. This is primarily because the structural characteristics of Chinese folk song variations differ significantly from those of pop music, explaining the failure of pop music structure analysis methods. We hope this work will inspire further research focusing on the structural aspects of Chinese folk songs.

Moreover, although SA-CNFolk also achieves the current state-of-the-art performance under the more stringent HR.5F metric, the notable gap between HR3F and HR.5F indicates that there remains room for improvement in the temporal precision of boundary predictions. In future work, we plan to incorporate more refined temporal modeling mechanisms to further enhance the accuracy and robustness of structural boundary prediction. We also hope that in-depth study of Chinese folk song variation structures will provide new insights and support for downstream tasks, such as structure-controlled music generation and emotion analysis closely related to musical structure.

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